EXPERIMENT 12

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Aim

To prepare an algorithm and test-case table, then write, compile, execute, and test a menu-driven C program using **switch-case** for operations on a positive integer.

Problem Statement

Input a positive integer and choose: **(1)** find the sum and product of its first and last digits, or **(2)** count digits lesser than, equal to, and greater than a given digit **k**.

Algorithm

1. Start and input positive integer n.
2. If n <= 0, display an error and stop.
3. Display the two menu choices and input choice.
4. Use switch(choice).
Case 1: last = n % 10. Copy n to temp and repeatedly divide temp by 10 until temp < 10; this is the first digit. Compute sum = first + last and product = first * last. Display them.
Case 2: Input k (0-9). Initialize lesser, equal and greater to 0. Extract each digit using temp % 10, compare it with k, increment the corresponding counter, and divide temp by 10. Display all counters.
Default: Display Invalid choice.
5. Stop.

Test Case Table

Test	Number	Choice	k	Expected Result
1	12345	1	-	First=1, Last=5, Sum=6, Product=5
2	9082	1	-	First=9, Last=2, Sum=11, Product=18
3	12345	2	3	Lesser=2, Equal=1, Greater=2
4	77707	2	7	Lesser=1, Equal=4, Greater=0
5	2468	2	5	Lesser=2, Equal=0, Greater=2
6	123	3	-	Invalid choice

C Program

```
#include <stdio.h>

int main()
{
    int n, temp, choice;
    int first, last, sum, product;
    int k, digit, lesser = 0, equal = 0, greater = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("\nMENU\n");
    printf("1. Sum and product of first and last digits\n");
    printf("2. Count digits lesser/equal/greater than k\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            last = n % 10;
            temp = n;
            while (temp >= 10)
                temp /= 10;
            first = temp;
            sum = first + last;
            product = first * last;

            printf("First digit = %d\n", first);
            printf("Last digit = %d\n", last);
            printf("Sum = %d\nProduct = %d\n", sum, product);
            break;

        case 2:
            printf("Enter digit k (0-9): ");
            scanf("%d", &k);
            if (k < 0 || k > 9) {
                printf("Invalid digit k.\n");
                break;
            }

            temp = n;
            while (temp > 0) {
                digit = temp % 10;
                if (digit < k) lesser++;
                else if (digit == k) equal++;
                else greater++;
                temp /= 10;
            }

            printf("Count_lesser = %d\n", lesser);
            printf("Count_equal = %d\n", equal);
            printf("Count_greater = %d\n", greater);
            break;

        default:
            printf("Invalid choice.\n");
    }
    return 0;
}
```

Compilation and Execution (Linux/GCC)

Compile: `gcc experiment12.c -o experiment12`

Execute: `./experiment12`

Sample Testing

Test 1: n = 12345, choice = 1 -> First = 1, Last = 5, Sum = 6, Product = 5. **PASS**
Test 2: n = 12345, choice = 2, k = 3 -> Lesser = 2, Equal = 1, Greater = 2. **PASS**
Test 3: n = 9082, choice = 1 -> Sum = 11, Product = 18. **PASS**

Result

The menu-driven C program was successfully prepared using switch-case and verified against the test cases. Both required operations produce the expected results.